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Education

University of Hawai‘i at Mānoa
PhD Candidate in Zoology
Advisor: Dr. Peter Marko

started August 2019
current GPA: 4.0

University of California, Los Angeles
Bachelors of Science Marine Biology
Minor in Environmental Systems and Society
Advisor: Dr. Paul Barber
Senior Honors Thesis: *Investigating Temporal Variation in a Marine Environment*

June 2018
GPA: 3.957

Publications

Fumo, J.T., Nichols, P.K., **Ely, T.**, Marko, P.B., Moran, A.L., Powell, B.S., ... & Sherwood, A.R. (2025). A predictive framework for identifying source populations of non-native marine macroalgae: *Chondria tumulosa* in the Pacific Ocean. *PeerJ*, 13, p.e19610.

Ely TD*, Mukai GNM*, Kobayashi DR, Marko PB, Moran AL, Wren JLK. (2024). Characterizing key sensitivities for modeling long-distance dispersal in the North Central Pacific Ocean. NOAA Technical Report. * = co-first authors

Ely T, Patten N, Naisbett-Jones LC, Spencer ET, Willette DA, Marko PB. (2023). Molecular identification of critically endangered European eels (*Anguilla anguilla*) in U.S. retail outlets. *PEERJ*.

Moran AL, Toh M-WA, Lobert GT, **Ely T**, Marko PB. (2021). Egg masses and larval development of the Antarctic cephalaspidean slug *Waegelea antarctica*, with notes on egg masses of the related *Antarctophilina alata*. *Journal of Molluscan Studies*. 87(3).

Ely T, Barber PH, Man L, Gold Z. (2021). Short-lived detection of an introduced vertebrate eDNA signal in a nearshore rocky reef environment. *PLOS ONE* 16(6): e0245314.

Blanchette A*, **Ely T***, Zeko A.*, Sura S, Turba R, Fong P. (2019). Damselfish *Stegastes nigricans* increase algal growth within their territories on shallow coral reefs likely via enhanced nutrient supplies. *Journal of Experimental Marine Biology and Ecology*. * = co-first authors

Grants

2024 to current - **Jessie D. Kay Memorial Award in Zoology** (\$10,000)
What are the dispersal pathways for marine species to the Hawaiian Archipelago?

2023 - **Elizabeth Alison Kay Endowed Awards** for students in Biology/Zoology at the University of Hawai‘i at Mānoa (\$2500)
Dispersal patterns of marine organisms in the Hawaiian Archipelago

2020 - **The Charles H. And Margaret B. Edmondson Research Fund** (\$2000)
Community-wide genetic patterns revealed through eDNA

2019 to 2024 - **NSF Graduate Research Fellowship Program** (\$118,500 over 3 years)
Using eDNA to examine the impact of warming oceans on marine fish communities

Experience

Graduate Research Assistant

September 2024 to current

University of Hawai'i at Mānoa

NSF grant # 2049672

Dr. Peter Marko's Lab

Investigating marine biogeography and population connectivity in North Central Pacific

- Collected and processed ~2,000 eDNA samples for biogeography and population genetics analyses
- Simulated larval dispersal into the Hawaiian Archipelago using python and OceanParcels

NSF Graduate Research Fellow

August 2019 to September 2024

University of Hawai'i at Mānoa

Testing environmental DNA as a population genetics tool in the Main Hawaiian Islands

- Processed samples for MiSeq; bioinformatics and non-parametric statistics in R
- Created functions and a population genetics pipeline in R using multi-species eDNA data

Lab Technician

February to June 2019

Bowdoin College, Brunswick, Maine

Sequenced and ran preliminary analysis on parrotfish gut samples

- Processing samples for MiSeq and qPCR
- Created a protocol for laboratory and bioinformatics analysis designed for undergraduate students
- Managed lab upkeep including ordering supplies

Oregon Sea Grant Summer Scholar

June to August 2018

Department of Fish and Wildlife (ODFW)

Marine Resources Program, Newport, OR

Assisted in the ecological monitoring of the intertidal communities within Oregon's marine reserves.

- Monitored sea star wasting syndrome (SSWS) and analyzed data from SSWS and biodiversity surveys in Oregon's marine reserves with both univariate and multivariate statistics
- Expanded the monitoring program through successful collaborations
- Led a team of 3-7 staff and volunteers during intertidal field sampling

Undergraduate Research Assistant

October 2015 to June 2018

UCLA, Department of Ecology and Evolutionary Biology, Los Angeles

Undergraduate in Dr. Paul Barber's Lab

Senior Honors Thesis: Investigating Temporal Variation in a Marine Environment

Invertebrate Zoology Intern

June to August 2017

Smithsonian Institute, National Museum of Natural History

Dr. Chris Meyer's Lab, Washington D.C.

Extracted tissue samples from ARMS across Indonesia and prepped samples for Sanger sequencing

Public Programs Intern

May 2015 to May 2016

Santa Monica Pier Aquarium, Santa Monica, CA

Assisted with outreach and events about aquarium animals and conservation issues

- Organized scientific talks including a talk with NASA JPL
- Created advertisements, social media posts, and displays using Canva
- Helped found the seabirds program

Teaching

Undergraduate Mentoring April 2025 to current
Advise a student working with eDNA data in R to investigate life history's influence on detections

Environmental DNA team leader July 2024
Larvae Across the Pacific Research Cruise *R/V Thompson*
Trained and supervised a team of undergraduates and post-baccs to process environmental DNA samples

Teaching Assistant January 2020 to May 2020
Invertebrate Biology (BIOL 485L) University of Hawai'i at Mānoa

Teaching Assistant August 2019 to December 2019
Ecology and Evolutionary Biology Lab (BIOL 265L) University of Hawai'i at Mānoa

Presentations

Oral Presentations

Ely T (2025) "Population Genetics of Marine Species in Hawai'i using eDNA". Albert L. Tester Memorial Symposium. Honolulu, HI.

Ely T (2025) "Analysis of eDNA as a community genetics tool" 2nd Australian & New Zealand Environmental DNA (eDNA) Conference. Wellington, New Zealand.

Ely T (2024) "eDNA: a promising tool for population genetics" Albert L. Tester Memorial Symposium. Honolulu, HI.

Ely T (2023) "Can eDNA be used as a population genetics tool for marine communities?" Albert L. Tester Memorial Symposium. Honolulu, HI.

Ely T (2022) "Testing eDNA as a fine-scale population genetics tool in marine ecosystems" Albert L. Tester Memorial Symposium. Honolulu, HI.

Ely T (2018). "Monitoring Sea Star Wasting in Oregon's Marine Reserves". Oregon Sea Grant Final Symposium. Newport, OR.

Ely T, Blanchette A, Zeko A (2017). "Territorial damselfish enhance nutrient supplies to algal turf in shallow coral reefs". Marine Biology Quarter Symposium. Los Angeles, CA.

Poster Presentations

Ely T (2018). "Monitoring Sea Star Wasting in Oregon's Marine Reserves". Oregon Sea Grant Final Symposium. Newport, OR.

Ely T (2018). "Investigating Temporal Variation of eDNA in a Marine Environment". UCLA's Annual Biology Research Symposium. Los Angeles, CA.

Ely T (2018). "Investigating Temporal Variation of eDNA in a Marine Environment". UCLA Research Poster Day. Los Angeles, CA.

Selected Graduate Courses

An Introduction to place-based methodologies in Hawai'i; People and the Sea; Population Genetics; Systematics and Phylogenetics; Biological and Physical Oceanography; Advanced Statistics in R

Certifications

- Scientific Diving certified since 2016: 112 scientific dives
- CPR, AED, and First Aid certified

Honors & Awards

2025	Honolulu ARCS Scholar (\$6,000)
2025	Honorable mention for graduate student talk at Tester Symposium
2025	Environmental DNA (eDNA) Conference Bursary (\$1500)
2018	Graduated Summa Cum Laude and Departmental Highest Honors
2018	3 rd place Undergraduate Poster at UCLA's Biology Research Symposium
2014-2018	Dean's Honors List
2017	UCLA Undergraduate Research Scholarship Program award (\$5000)
2017	Whitcome Research Undergraduate Summer Fellowship (\$3000)
2016	Promise Award Scholarship (\$2000)
2014	California State PTA Scholarship (\$500)

Community Service

Various workdays on O'ahu	2024 - current
Paepae o He'eia; Ka Popo Lo'i o Kānewai; Waiale'e Lake Pono; 808 cleanups; Kaiāulu 'O Kahalu'u,...	
International Save the Vaquita Day	September 2024
Lead volunteer, ran an educational booth (reached 85 visitors)	Waikīkī Aquarium
STEM outreach for kids	March 2020; October 2022; February 2023
Kupu Hawai'i STEM outreach; Career Day at Dreamhouse HS; The Science of the Seuss Library Event	
Leeward Science and Engineering Fair	February 2020 and 2021
Volunteer judge for environmental science and biological sciences categories	online
Environmental Message of the Month	September 2013 to June 2014
Founded and ran this program to educate students and parents	Palos Verdes Intermediate School
http://monthlyenvironmentalmessage.weebly.com/	

News & Media

UH News Article	September 2024
https://www.hawaii.edu/news/2024/09/08/research-voyage-marine-links-north-pacific/	

Professional Service

Student Social Chair	Spring 2025 to current
Co-lead of weekly Pau Hana event for graduate students and faculty	University of Hawai'i at Mānoa
Navigating the Publishing Landscape Workshop	October 2024
Discussion panelist for graduate student perspective	University of Hawai'i at Mānoa

Reviewer for Journals

Aquatic Conservation: Marine and Freshwater Ecosystems
Proceedings of the Royal Society B: Biological Sciences
Molecular Ecology Resources